AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

INDE 301
Engineering Economy

Term: Fall 2026 – 27

Class Time: 10:00 → 10:50 AM (IOEC – 210B)

AMERICAN UNIVERSITY OF BEIRUT

Instructor

- Naim Al Khoury (na381@aub.edu.lb)
  - Mechanical Engineer
  - Sanita – INDEVCO Group

    Production planning, inventory management, product development maintenance management, process control, project management, …
  - PhD in Industrial Engineering and Operations Research @ Ghent University

    Data-driven inventory management under uncertainty
- Office Hours:
  - MWF 11:00 → 12:00 at RGB – 406
  - Appointment

Why Industrial Engineering

- https://app.wooclap.com/events/GFUUESQ/live-session

Why Engineering Economy?

- Industry need:
  - Engineers with strong technical background to design technical solution(s)
  - These solution(s) must make sense financially

The economic evaluation of alternatives to allow for informed decision-making

Example

A local bakery needs a new van to deliver fresh bread.

- Option A: a used gas van that is cheap to buy but expensive to fuel.
- Option B: a new electric van that is expensive to buy but cheap to charge.

Which vehicle is the smartest financial choice over the next five years?

Engineering economy provides the exact tools to compare these alternatives.

Why Engineering Economy?

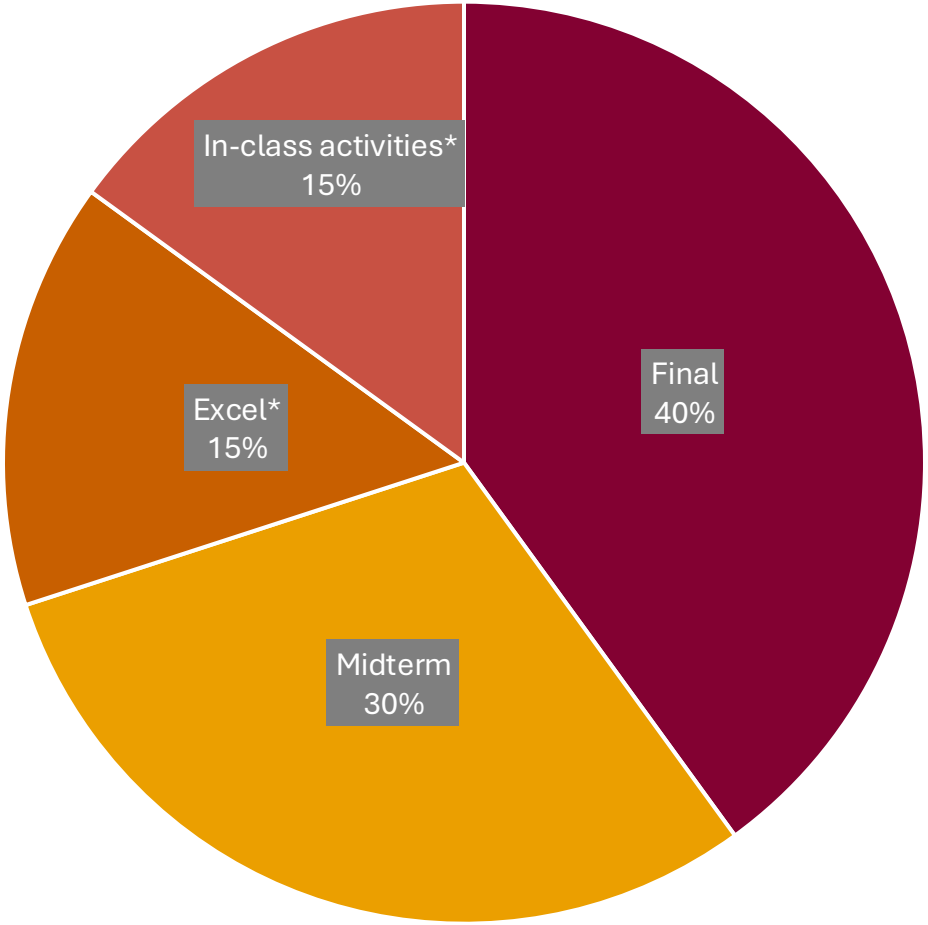
- Not only industry, the concepts and tools are applicable to your everyday decision making:
  - Buy an expensive car vs a budget car and invest remaining
  - Eat everyday in fancy restaurants vs save the money and dwell into a new business venture
  - Should I replace my phone or wait till next year
  - ⋮

Course Organization

- Theory Lectures + course notes
- In-class activities (teams)
  - Attendance
  - Paper based solving
  - Small assessments – frequent and unannounced
    $R = (best\ 6)\ - (incomplete\ activity - 2)^{+}\ \times\ 2.5$
  - Excel (bring laptops/tablets)
- Homework – not graded



Grading Scheme



*Tentative

Why Excel

- Excel dominates, so far, the workspace
- 1.1 – 1.5 Billion users
- Top companies still require advanced Excel for Financial Planning & Analysis / financial-analyst roles
- > 80% of engineering teams rely on Excel for design calculations
- 30% use Excel for engineering-to-order / configure-price-quote workflows

Some Basics

- Maturity
  What you do now will shape your future career

- Attendance
  - Interaction in class is important
  - Everyone of us gets sick, …

- Artificial Intelligence
  - Encouraged to **support** you, but not to **replace** you
  - Be honest and scientific about its use
  - Do not generate excel, reports, or presentations using AI

Questions?

